



Building Custom Fleet Management Software:

Complete Guide with an Emphasis on Emerging Technologies

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Concluding remarks...

Introduction

What are the trending technologies for fleet management?

How to build a fleet management system incorporating these tech?

What's the tech stack going to look like?

How will this system address the existing pain points of the fleet?

As a technology leader, it's obvious for these questions to zoom across your mind. With the post pandemic norms and increasing complexity of fleet management issues, SaaS offerings don't suffice anymore. In such a scenario, it makes perfect sense for CTOs to rely on prevalent and futuristic technologies to accelerate their fleet's growth.

With the help of this ebook, you'll be able to look beyond canned solutions. You'll get to know all about emerging tech, ripe-enough for you to include in a fleet management software, meticulously customized for your fleet. We will introduce you to pivotal technologies for fleet management, discuss their applications, and guide you through the development process.

Introduction to Fleet Management and its Challenges

Businesses of all sizes and nature often need a certain number of vehicles at their disposal to carry out day-to-day activities. It could be a manufacturer using trucks to move finished products to sellers or a law firm having multiple small vehicles for their executives. If your business is managing more vehicles than you can count on your fingertips, fleet management becomes critical.

Overseeing vehicle movement, maintenance, driver management, and regulatory compliance are some tasks that fall under the umbrella of fleet management. Apart from these, you also expect fleet managers to find new ways to improve productivity, work on driver safety, optimize vehicle performance, and cut down costs.

Understandably, managing these tasks gets increasingly challenging with addition in the fleet's size and operation complexity. Your fleet manager can only take care of so many things at a time, and stiff competition doesn't favor them either. Thankfully, technological interventions have helped fleet managers and businesses address this issue. They have had a profound impact on fleet management over the years and continue to do so even today.

In this chapter, we'll discuss challenges that businesses face while managing their fleet. We will also touch upon how modern technologies help alleviate those challenges and make fleet management a breeze. So let's dive right into it:

1. Operational Efficacy

Fleets are resource intensive. Right from heavy-duty trucks to the skilled workforce, everything that goes into it requires businesses to shell out big bucks. So it's only obvious that you'd want to make the most of your on-hand resources.

We've come through the era of spare units. Companies have pretty dramatically rationalized their fleets so that all units are working every day.

Dave Johanneson

Group VP for Navistar International Transportation Corp.'s truck business, told [FleetOwner](#)



So how to get fleet managers to bring in more efficiency and operate lean?

- Use IoT and telematics to keep them informed about everything happening with the fleet at all times.
- Use Big Data and AI for better decision-making and even automating some of them to reduce a fleet manager's burden.



2. Optimize Fuel Expenditure

Fuel is the [second](#) most expensive thing in fleet management. Fluctuating fuel prices, crowded routes, rough driving, and low-performing vehicles are factors contributing to higher fuel consumption. Carlsberg supposedly saved [502,549](#) liters of fuel, amounting to nearly \$740,000 by enforcing eco-driving techniques. You can also keep the lid on fuel expenditure by placing similar strategies in place. Here's how emerging technologies can help you in this endeavor:



Determine optimal routes for your vehicles with Big Data and ML.



Keep a check on aggressive driving behaviors with telematics.



Keep the vehicles well-maintained with the help of predictive maintenance.

3. Improve Driver Safety

Driver safety should always be among the top priorities of your fleet manager. Apart from the obvious benefit of keeping the driver safe, this practice preserves the company resources, avoids fines, and builds a great rep.

Analyzing driver behavior is essential for fleet managers as well as the drivers themselves. As fleet managers are not physically in the cab with their drivers, telematics can help to ensure that a company's policies relating to driving behavior, are being followed

Scott Sutarik

Associate VP of Commercial Vehicle Solutions at Geotab while speaking with [asmag](#).

Here's how tech interventions improve driver safety:

- Use AR for driver training and placing the right drivers behind the wheel.
- Equip vehicles with IoT devices to aid drivers and keep fleet managers informed of all on-the-road updates.
- Use data analytics to generate comprehensive driver-performance reports.



4. Predict ETAs Accurately

With more businesses moving towards just-in-time inventory management, fleets need to be accurate in assigning delivery times. Your fleet managers need to take into account route conditions, the freight's nature, and some other factors to ensure the timely delivery of a dispatch.

Modern fleet-management software allows them to better manage many of these variables and stay on schedule in most cases.

Here's how it happens:

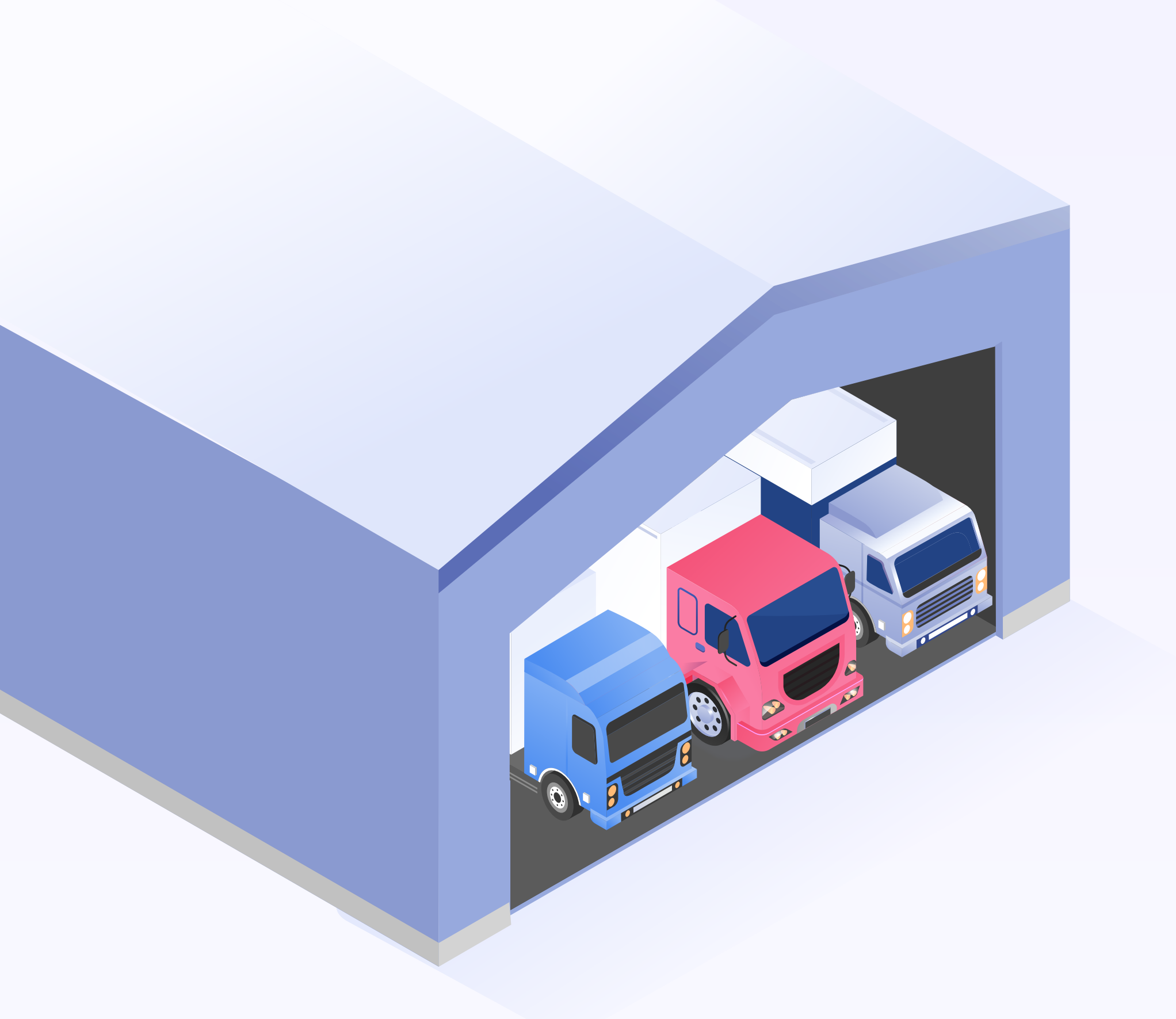
- Data analytics determines the optimum route for smooth and hassle-free transportation.
- Telematics handles unexpected situations such as blocked roads or vehicle failure very well.

5. Regulatory Compliance

Laws have modified over the years to ensure safe and greener operation of fleets. However, their compliance can be especially challenging for fleets operating across state and international borders owing to varying laws. Naturally, remembering every rule would be quite beyond a driver's pay grade. So it pays to have systems that make quick work of countless compliances.

This is how modern fleets are ensuring compliance:

- Using ELDs to provide authorities with reliable information about fleet operations
- Placing a mechanism in place that notifies the fleet manager of any violations in real-time



6. Driver Shortage

The American Trucking Association has estimated a shortage of **175,000** truck drivers in the US by 2026. It's a rare problem of the fleet industry that's escalating with time. Skilled truck drivers aren't easy to come by, and it's highly likely that your competitors might try to poach your competent drivers.

So the fleet managers often need to operate with the limited number of available drivers whilst also ensuring perfect working conditions for higher retention.

This is how modern fleet-management solutions deal with the shortage:

- Semi-autonomous vehicles are clearly the answer to the problem.
- Advanced analytics deploys drivers in the most optimum way possible.
- ELDs make sure the drivers aren't excessively strained.
- AR solutions augment the driver's capabilities.

7. Vehicle Maintenance

You need to have a robust system for top-notch maintenance to the backbone of your business – the fleet. Vehicle failure during operations not only adds on to the overall cost of maintenance but also hurting your reputations because of delays.

Moreover, maintenance should also be smooth and less time consuming. This leads to more vehicles on the road and fewer in the hub. Several modern technologies come together for predictive maintenance:

- Telematics notifies about upcoming maintenance events.
- Wearable AR tech makes inspections a cakewalk and speeds up the process exponentially.
- AI brings in predictive maintenance and schedules checkups if a vehicle isn't handled correctly on a particular trip.



8. Integrating FMS With Other Business Software

Fleet-management software isn't going to be the only tool you use to facilitate your business operations. CRM, ERP, accounting software, and HR are some tools used for aspects other than fleet management. However, it's desirable that the FMS you deploy syncs with these tools for efficient functioning.

Higher integrability leads to quicker adoption and implementation, smoother workflow, and improved communication between various departments. For integrations, Simform has got you covered:

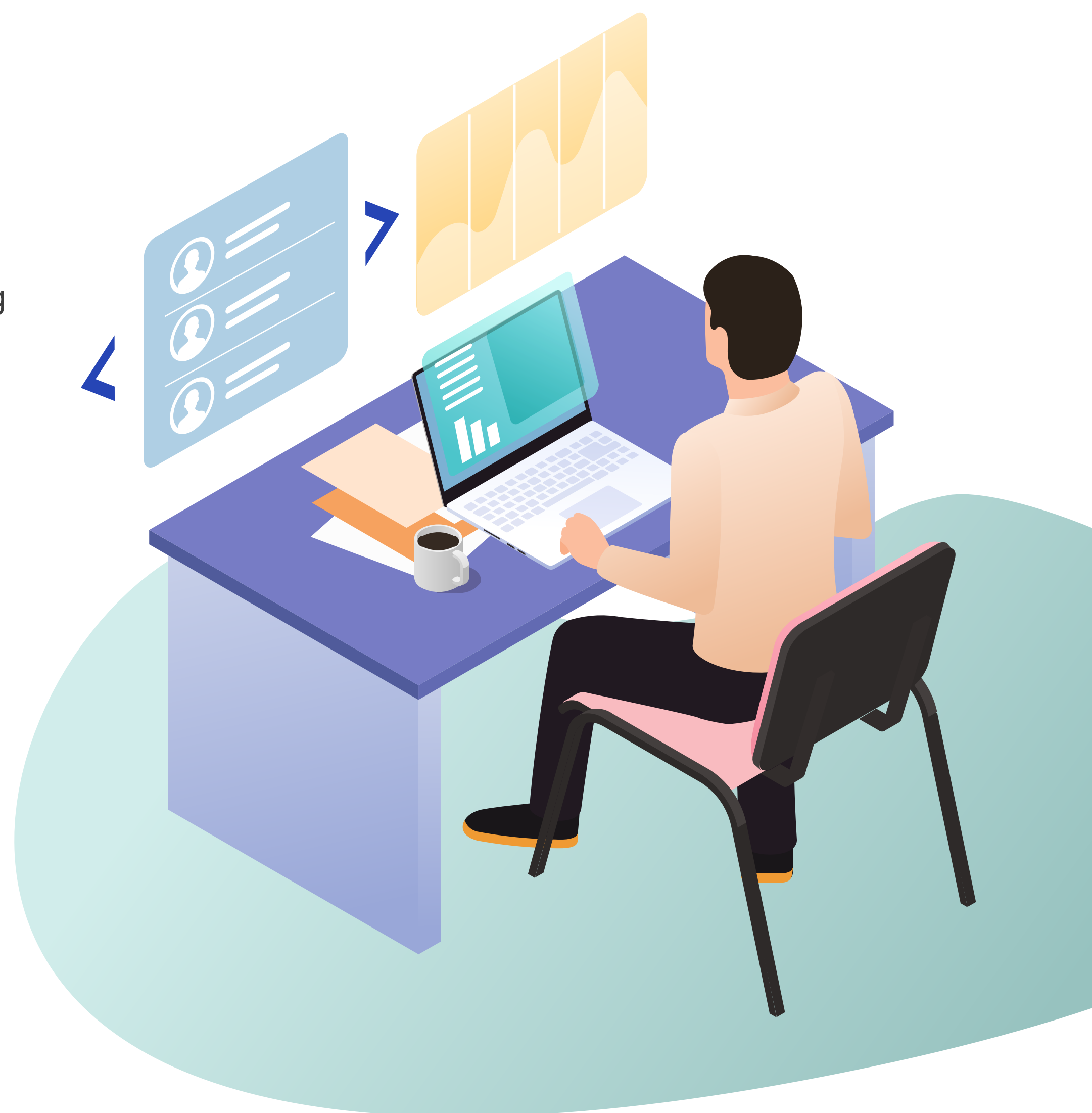
- With our API integration expertise, you can easily harness the combined power of the different software you use.
- Our talented developers will build custom APIs that improve the functionality of applications you use.

9. Speed up Admin Tasks

Your fleet managers often end up spending hours on unproductive back-office operations. Whether it's filling in reports, invoicing, or taking care of the payroll based on tasks, they can all be automated. And your fleet manager could focus on more productive activities instead.

There are many ways an FMS can relieve them of such tasks:

- It can generate error-free reports with deep insights into operations.
- It dispatches vehicles and delegates tasks to available drivers.
- ML algorithms can go through documents and analyze them to filter out any unwanted or potentially dangerous terms.



10. Implying Green Initiatives

Governments, businesses, and even customers are now moving towards greener options. Fleets across the globe are also paying close attention to their carbon footprint. However, becoming fully environment friendly is a long, tough road ahead.. So as a responsible business owner, you need to start with smaller steps towards a greener fleet.

There are various fronts on which you can make your business more environmentally conscious, and tech will always be around to support the cause.

Here's how:

- Telematics reduces vehicle-idling cutting down on a lot of unnecessary emissions.
- Digitization will make your business go paperless.
- AI and ML encourage better driving behavior, which incurs less stress on the environment.



Modern solutions can deal with all these challenges you come across with modern fleet management. However, it won't be a smart approach to tackle these issues separately. Instead, you want a dynamic tool capable of addressing them all at once. So yes! We're talking about fleet management software although not all of them are the same.

You need to be mindful of your fleet's pain points and need to pick a software that can deal with them. In the next chapter, we'll discuss the qualities in a desirable FMS, which will allow you to choose the tool best suited to your needs.



Must-have Features in Fleet Management Software

We spoke about new technologies pushing the envelope for fleet management software. We also learned how these technological advancements translate into an improved bottom line for your business. But does that mean you need all available features in your FMS – especially when they can potentially burn a hole in your pocket?

Well, of course not! Only for the rarest of use cases will you be required to cover the entire spectrum of these newer technological applications. For everything else, a select handful of essential features can easily see you through most fleet-management challenges. Let's have a look at the features your FMS can't do without:

1. Driver Management

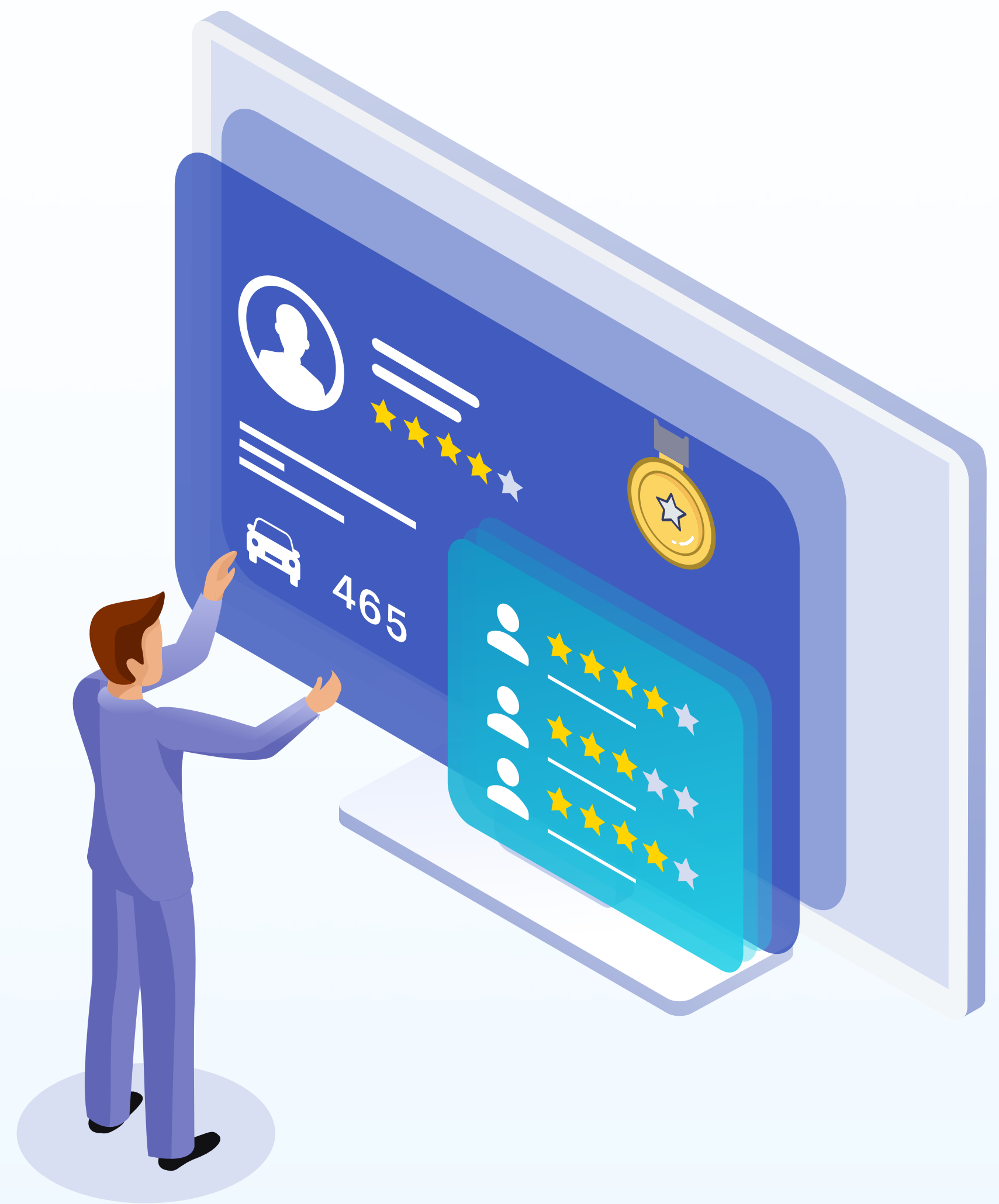
This feature encapsulates a lot more than just **assigning drivers** to specific trips. You can generate **comprehensive driver profiles** based on their **past behaviors** and **current driving habits**. These profiles make it easier for you to identify truly valuable assets for your organization and **reward** their appreciable efforts.

The most crucial element of any fleet is, no doubt, the drivers responsible for delivering on time and taking care of the vehicle. With the driver-management feature, you get superior control over this key element to elevate the efficiency of most fleet operations.

2. Routing and Tracking

While in transit, the asset status is among the biggest concerns of fleet managers and customers alike. Thankfully, the routing-and-tracking feature takes care of it from A to Z. It starts with **planning the optimum route** while taking all the variables and the deadline into consideration. And once the cargo is on the road, it keeps you posted with timely **alerts and notifications**.

The feature also brings you **real-time ETAs**, which massively improves the reliability of your business. The continuous **asset tracking** brings great transparency into fleets and more power to your fleet managers.



3. Fuel Management

The importance of efficient fuel management in fleets could never be stressed enough. This feature helps you stick to the **fuel consumption targets** to reduce your fleet's carbon footprint and restrict avoidable expense. It gives you deeper insight into what driving behaviors save more fuel and prompts encouragement for similar actions across the fleet.

Your fleet managers are also able to better **monitor and control fuel purchases** remotely. With this feature, they can accurately **predict the number of refills** needed on the entire route and coordinate it with the availability of fuel stations along the way.

4. Predictive Maintenance

Here's a big one! This feature kicks guesswork out of the equation, and your vehicles go to the yard for maintenance only when needed. **On-board diagnostics reports** give you reliable reports on various parts of vehicles and help extend their life and **ROI for your business**.

Your managers also get **alerts for wear and tear** and can easily direct drivers to get the issue resolved and to prevent any further damage. The feature is also capable of **leveraging the service history** of the vehicle to **schedule repairs**.



5. Fleet Safety and Compliance

The driver's safety and health surpass all other concerns for fleet managers. The feature of fleet safety and compliance ensures that your **drivers operate within regulatory restrictions**. It keeps your driver as well as the cargo safe while on the move.



The feature is also great for **disaster mitigation**. It **automatically deploys resources** for help if a driver is in distress. And once normalcy resumes, your managers can go through **reports** to determine the root cause of an unwanted situation. Plus, the feature saves your business from unnecessary fines and disputes with authorities.

 Cargo safety with installed sensor	 Get instant notification and alert in case of hazards	 Get notified on regulatory violations to avoid fines.	 Rested for optimum performance	 Automatic invoicing & billing
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The features above are at the core of any modern fleet-management software and take care of most challenges thrown at your fleet manager. You can't afford to overlook them if you plan on getting management software for your fleet.

Speaking of getting an FMS, the next chapter discusses how you could build one tailored to your business goals and needs. Keep reading to learn more.

How to Develop a Fleet Management Software

With the evolution of digital solutions and software-enabled objects, almost every business has made its internal and external components more efficient and productive. With the fleet and logistics industry, the scenario is no different. It's come to be dominated by software as a service domain where digital programs and satellite connectivities manage fleet operations. Challenges in keeping track of fleet operations are common across many organisations. Keeping a record of logistics, drivers and cargo-vehicle allocation, and other related tasks demand an analytical solution. The aim is to minimize resource consumption and at the time be as efficiently as possible for internal profits and clients.

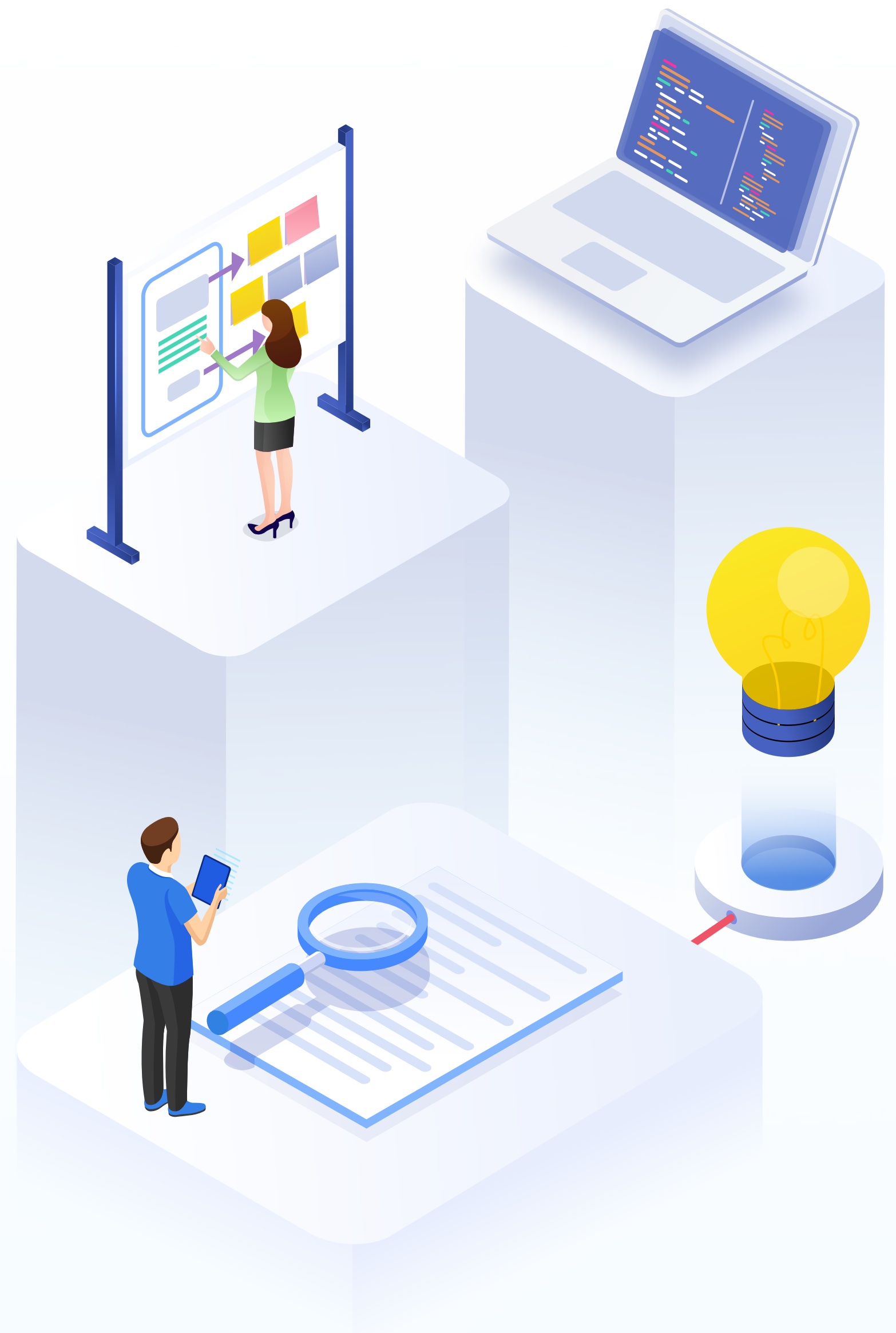
Though several organizations feel satisfied with internal resources, manual backlog often throws a wrench in smooth functioning of daily tasks.. Companies have also employed hiring in-house resources to build software applications. However, the process is so technically dense and challenging that anything short of a dedicated team of experts just doesn't cut it.

This chapter will discuss some crucial aspects of building a custom fleet-management system. So how to build custom software in fleet management? Or what would you need to check before opting for one?

Increasing Productivity in Fleet Management With Custom Software

As a business owner, you must've considered many options that could actualize the need of business-automation. But understanding the difference between developing a custom software system and opting for a ready-made one is important.

We're going to recount a few features to help you understand the benefits of custom software in fleet management:



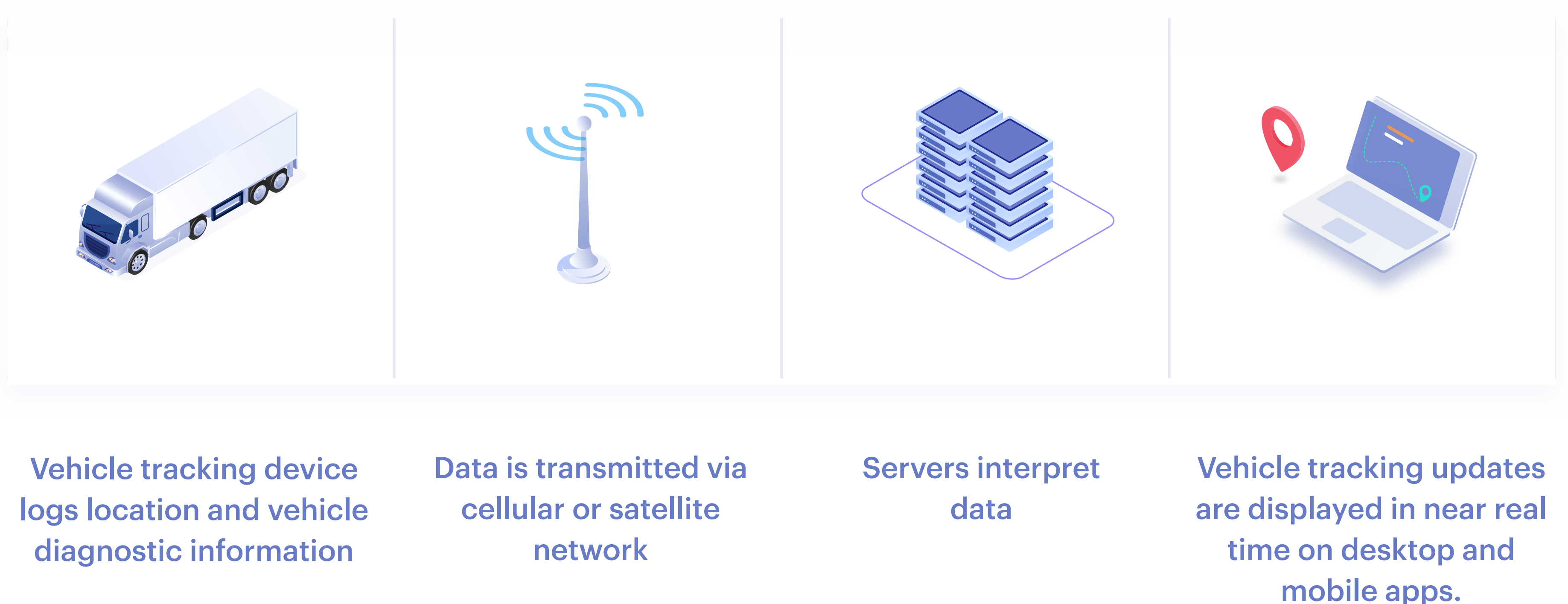
Driver & Vehicle Oriented Customization

Custom-software systems are made to fill in the gaps of business processes that otherwise chip away at the profits. Your choice of software should ensure drivers' safety on the road, manage fuel consumption and excess insurance, and optimize vehicle downtime. It should also take into account taxes and probable penalties ahead of the journey!

What is Telematics?

Telematics combines the technology of communication with that of information gathering and reporting. Your vehicle would carry the software/intelligent system that could receive detailed information about your vehicle like speed, idle time, fuel consumption, tire pressure, and more. It should also be able to send optimization feedback based on these factors.

So the purpose of this mechanism is to get real-time information. For fleet and logistics business, there are a number of trucking companies for hire that deal with multiple vehicles travelling across different regions of a country and continents on a daily basis. To centralize the business and prevent any misguidance, an accurate software system helps you get necessary details on screen. Only, the information has to be accurate, readable and quick.

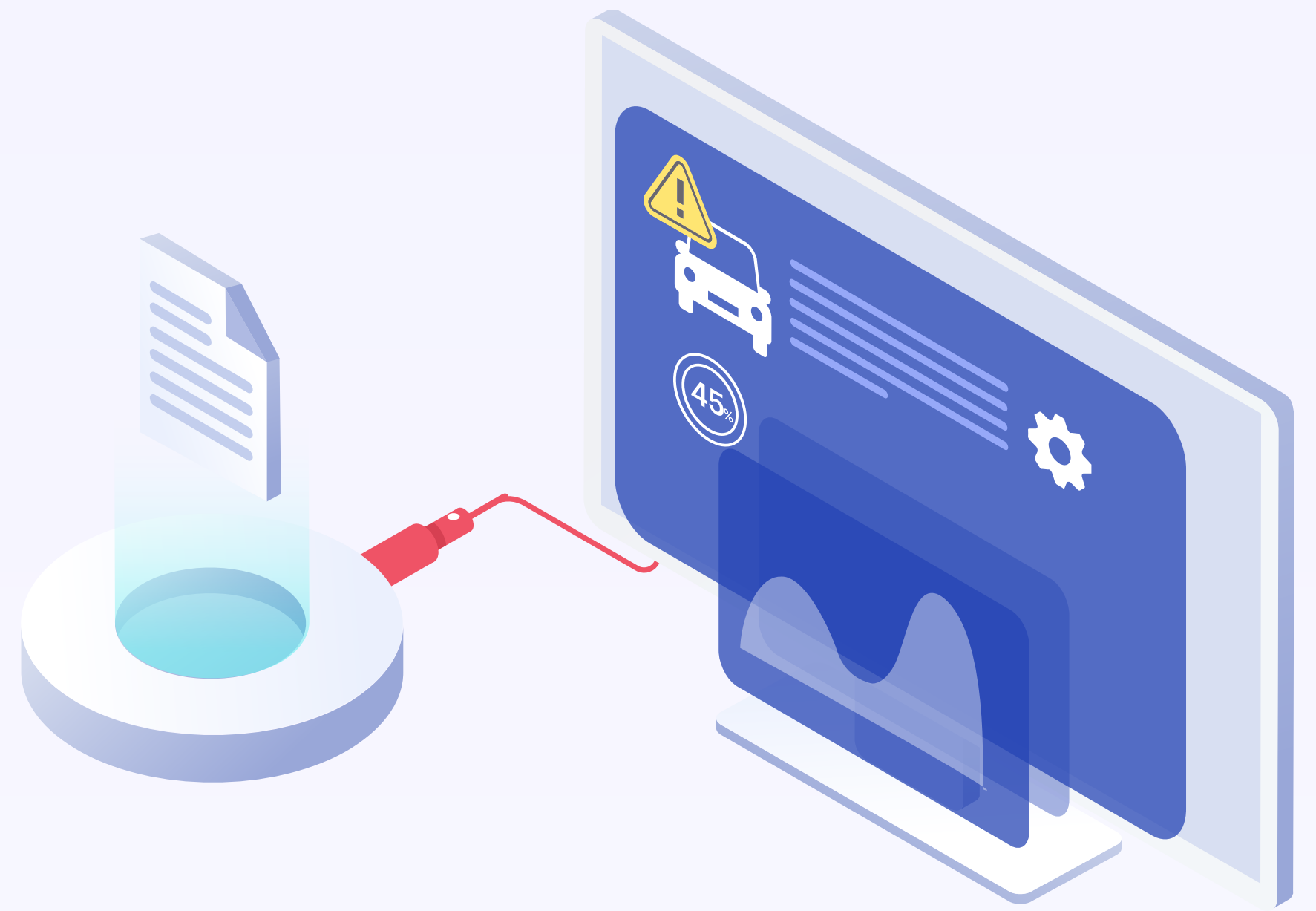


How to Start Customizing on Fleet Management Software?

To What Extent can You Map Vehicle Behavior at a Given Location?

The aim is to collect accurate information so that fleet managers can focus more on organizing business productively. It helps business owners speed up deployment during peak hours and address critical issues in real-time. Digitally-enabled solutions not only bring productive results but are also future-proof.

For example, the custom fleet-management software is now integrated with the **Internet of things**. This collaboration brings a whole new perspective of producing cost-effective solutions and building asset-tracking systems to fetch the information that fleet managers need and compartmentalize what's important. These systems are intelligent and more focused on connectivity. The Internet of things is about connecting every object through the internet and uses fleet software to generate reports, to monitor and give commands, and to produce information in a readable format for us.



Customization for Tracking and Alerts

Once desired telematics is in place, customized alerts and tracking assets become more feasible. You could bring a revolutionary change to your transportation system and handling of business. With the advent of technologies like the Internet of things, you could get specific alerts and push notifications via the asset tracking device, which help make critical decisions. In case of emergency or real-time updates, you can easily set these alerts configured with mobile devices and get the ground-level staff to act immediately.

Sometimes, temperatures, climate, unpredictable circumstances like national emergencies, or even the vehicle's central temperature experiencing malfunction can affect the transportation systems. In such cases, you'd be able to get quick updates and take measures to overcome the temporary snag.

With the Internet of things, organizations are more equipped to track the functioning of a vehicle. You can customize your fleet management software. This way, you can connect with several devices and have a centralized control over fleets running over different locations. When logistics business units need to take care of their warehouses, fixed assets or idle vehicles, geofencing is a great aid.

With geofencing, you could track stolen idle assets or check the location of certain assets at any given time while organizing the schedule for multiple fleets. It saves a ton of resources, avoids misallocation of critical equipment including vehicles, and speeds up the transportation process.

Tracking Driver's Activity and Addressing Security Concerns

Managing fleets is difficult to say the least, especially over larger regions. So it's understandable that fleet managers would need to monitor fleets right from their desks.. But without any automation help, trying to achieve this manually is a nightmare. Thankfully, custom FM software should monitor it all, so you can better focus on allocating resources, optimizing cost, assisting in route management, and decreasing the turnaround time from the delivery, among other priorities.

Fleet-management is all about how you plan and schedule the trips and fulfill the delivery demands. Adding FM software to your organization would equip you to make better decisions, elevate your customer service, and optimize fuel consumption.

Moreover, choose a custom software that identifies the availability of drivers while also giving you the information about their location and performance. It should also enhance a driver's visibility and pinpoint the factors that could boost their performance by eliminating frequently-encountered challenges.

Keeping driver's safety in mind, the FM software should provide insurance access and notifications, vehicle downtimes, and proactive reminders for penalties and fines on certain routes and locations.

On the other hand, telematics ought to include safety measures such as seatbelt, speed limit, continuous communication while driving, and many others for a safe journey. The aim is to streamline the communication system between management and end workers, i.e., drivers and helpers.

The in-vehicle reporting and productivity of drivers directly influence business revenue. The system also includes a calling facility to stay in touch with the drivers while on trip. This prompts drivers to take necessary breaks and adjust hours accordingly to complete the required time shifts. Eventually, it saves time without exerting too much strain on the drivers.

Vehicle Maintenance by FM Software

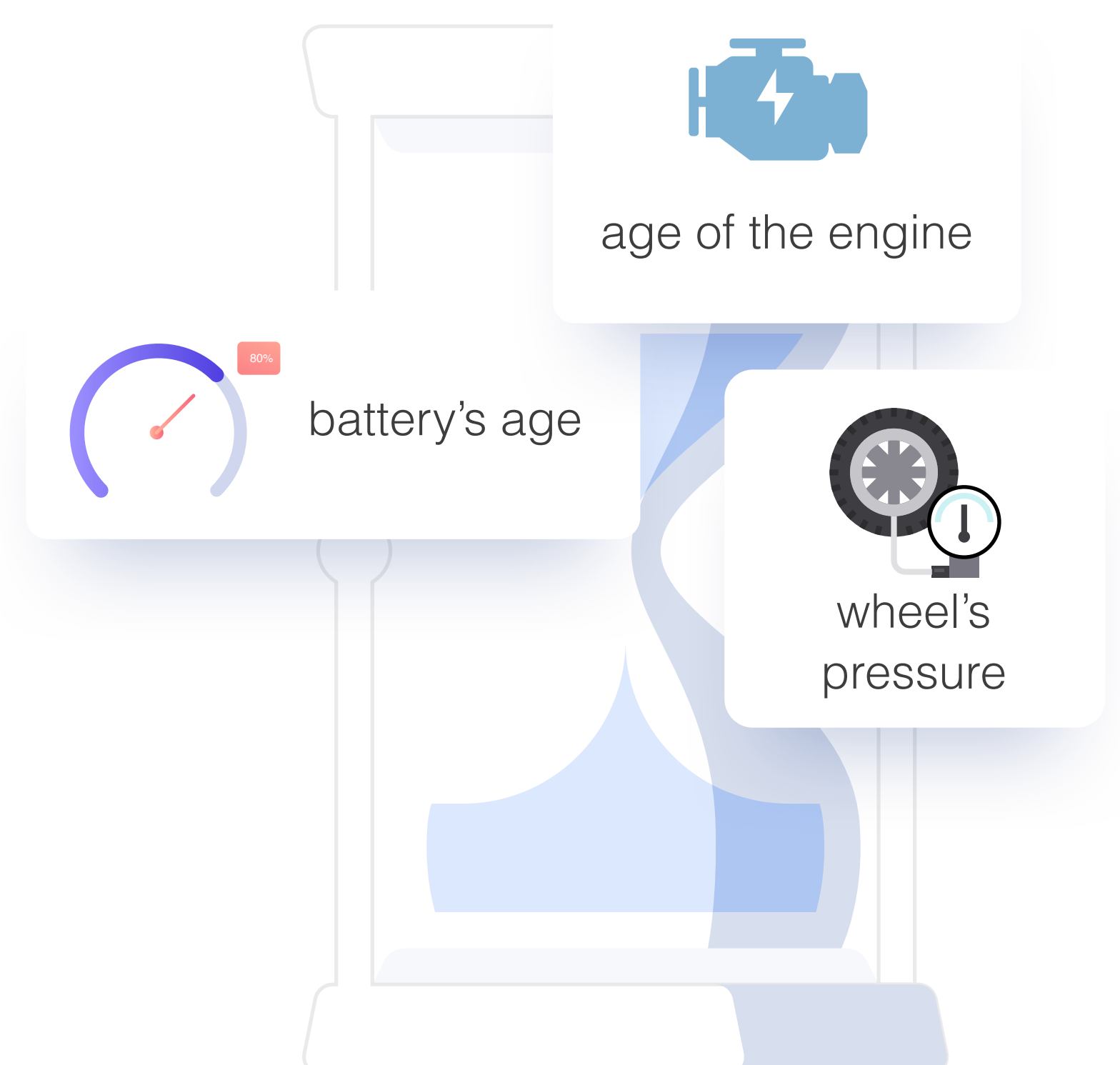
A telematics-based software can track the usage of vehicles to schedule advanced maintenance of certain types. It keeps fleets from experiencing unprecedented breakdowns and the subsequent failure of not making the delivery on time. FM software assists fleet managers in keeping tab on recovery of vehicle warranties, timely services, battery's age, temperature, wheel's pressure, age of the engine and other parts of the vehicle, and more.

This way, a continuous diagnosis of vehicles/fleets is received – decreasing overall maintenance and sudden costs.

This software is essential to eCommerce businesses. They have everyday deliveries and committed scheduling of products, where proactive maintenance comes as a life-saving add-on. Avoiding unpredictable maintenance would also mean **saving productive hours** and **increasing efficiency** of the business.

This intelligent, IoT-based software calculates all the related aspects of your vehicle, reflects them on connected devices and keep people and process very well informed.

Still, it does take a lot of effort to analyze aspects such as the exact hours or kilometers a vehicle has run. Using the technology, you can take preventive measures to reduce breakdowns, idle time, and allocation of resources and assets.



Accident Tracking via Mobile App

Accidents are a major concern when designing any vehicle, working on road construction, or taking a new route. Governments need to safeguard its citizens through the most effective measures and put maximum effort into enhancing people's safety. Not to mention how expensive and time-consuming it is to repair certain vehicles at specific times and locations, tow the car to the repairing services, or get an ambulance in remote areas. In such cases, a smartphone tracking system comes to the rescue.

Smart fleet-management systems use the global positioning system(GPS) to leverage the intelligence of the software. It monitors drivers' speeding, braking, and overall driving behavior. These systems use different phases and maintain connectivity to mobile devices throughout the journey. They carry embedded systems like smartphone's accelerometer sensors. This embedded system aims to extract vehicles data like speed, conditions of a reported collision, and other important details.

In case of accidents, smartphones are considered part of the car, so the smartphone app transfers the same experience to the particular system in detailed reporting forms. Then, the information is transmitted via notification to the reporting system/device with complete accuracy.

With the help of the GPS, location of the incident, time of the accident and movements of the vehicle can be relayed to the immediate responders. The smartphone accident-tracking system should also be capable of recording video, mapping the speed of the car/vehicle, and successfully deploying airbags. These essential details go on to decide the severity of the accident.

On-Road Assistance for Fleets

It is observed that drivers are most likely to seek guidance and monitoring at remote areas. The tracking system would automate on-road assistance for these drivers and fleets in need.

For example,

- A 24/7 helpline service
- The ability to resume driving as soon as possible
- A quick on-spot assistance for all types of vehicles including commercial trucks, fleets, heavy duty vehicles and more
- Seamless communication between immediate responders and drivers to report any unexpected problems
- Easy mobile-app access to towing, locksmith help, battery maintenance, and other services



Budget Constraints

Fleet-management software comes in all forms and formats, and there's something for every level of management of your fleet. So you need to know which available services you'd require for your business.

When you think of fleet management software, it seems that it's all about tracking the vehicles and adhering to the electronic logs, and capturing the analytics in some applications. While the entire concept has a comprehensive program and based on that, the budget is estimated.

The critical factors while estimating the budget includes mobile apps in connected vehicles, WiFi hotspots, temperature monitoring, sensors, IFTA reporting, RFID tags, and other technical aspects. If you choose to develop software from the ground up, it takes another estimation of pricing. Simultaneously, there are also successful case studies of businesses that integrate IoT and other advanced analytics in the existing software system. Both could have different budget perspectives.

Custom fleet management software includes a complete end-to-end business scenario to be looked up to. It's all about your physical assets, predictive maintenance, hardware embedment, reporting and analysis, driver allocation and management, and more. You can scale the software system as much as you require based on the budgetary elements. However, consulting reliable software development partners always prove to be a critical part of business uplifting and expense optimization because of their best-developed practices and experience.



IoT & Cloud in Fleet management

For industries like transport and logistics, IoT and machine learning solve many complex business problems, improve customer service, enhance productivity, and upscale the techniques of monitoring drivers and assets. IoT, with the help of cloud computing, makes efficient software solutions that increase the effectiveness of business tasks, eliminate tedious manual tasks, and bring the best results with higher productivity.

For example, traditional trucks are now replaced by smart trucks, whose core concept is to sync with techs like the Internet of Things, Machine Learning(ML), and Artificial Intelligence(AI) using integrated software solutions. In the coming years, IoT and machine learning are set to transform the business ways of the fleet industry.

Business Prospects of Fleet Management Using IoT and Cloud Computing

Customized mobile apps or custom fleet-management software make business operations very easy. A telematics-enabled software should work for all your business needs such as accurate tracking of the entire fleet, asset tracking, weather conditions, driving-pattern assessment, and extraction of specific information from raw data.

These IoT-integrated custom applications require a cloud connection to transfer high-level data. For instance, fleet operations are often complicated with variables like multiple locations, numerous resources, workforce allocation, vehicle diagnostics, and preparing reports. All these tasks generate a lot of data that need high-speed internet and fool-proof accuracy.



Solutions to Fleet Challenges Through Cloud Computing & IoT

Cloud computing enables fleet managers to have centralized control over the arrival and departure of fleets, monitoring different transportation methods, easy sharing of data using software-based sensors, and examining business-activity patterns.

You can eliminate the need for physical infrastructure of servers with this on-demand availability of systems. Reducing cost and time of operations, it lets you access virtual machines that are better equipped to handle massive data storage, data processing, and necessary computing.. This way, business processes are monitored automatically without human intervention. You can review the results and analytics via the mobile app, custom software, and remote dashboard.

Moreover, traditional limitations of infrastructure like physical servers failing to fully serve increasing users and handle colossal data transfer are no longer a deterrent. With cloud technology, your business will have unlimited access to this flexibility of scaling the functionalities of your application. You don't want to opt for a complex licensing process while already getting huge benefits from connected devices, high-speed internet connectivity, and establishing seamless integration between the various verticals of your business.

Furthermore, when you have enormous data from different fleet locations, assets, and resource allocation, it becomes challenging for fleet admins to break down this raw data into valuable nuggets of information and derive business benefits from it. For this reason, you need to ask your app developers or outsourcing partners to opt for on-demand cloud-computing platforms that convert data into valuable insights with options like [AWS IoT Greengrass](#).

IoT & Supply Chain Depots

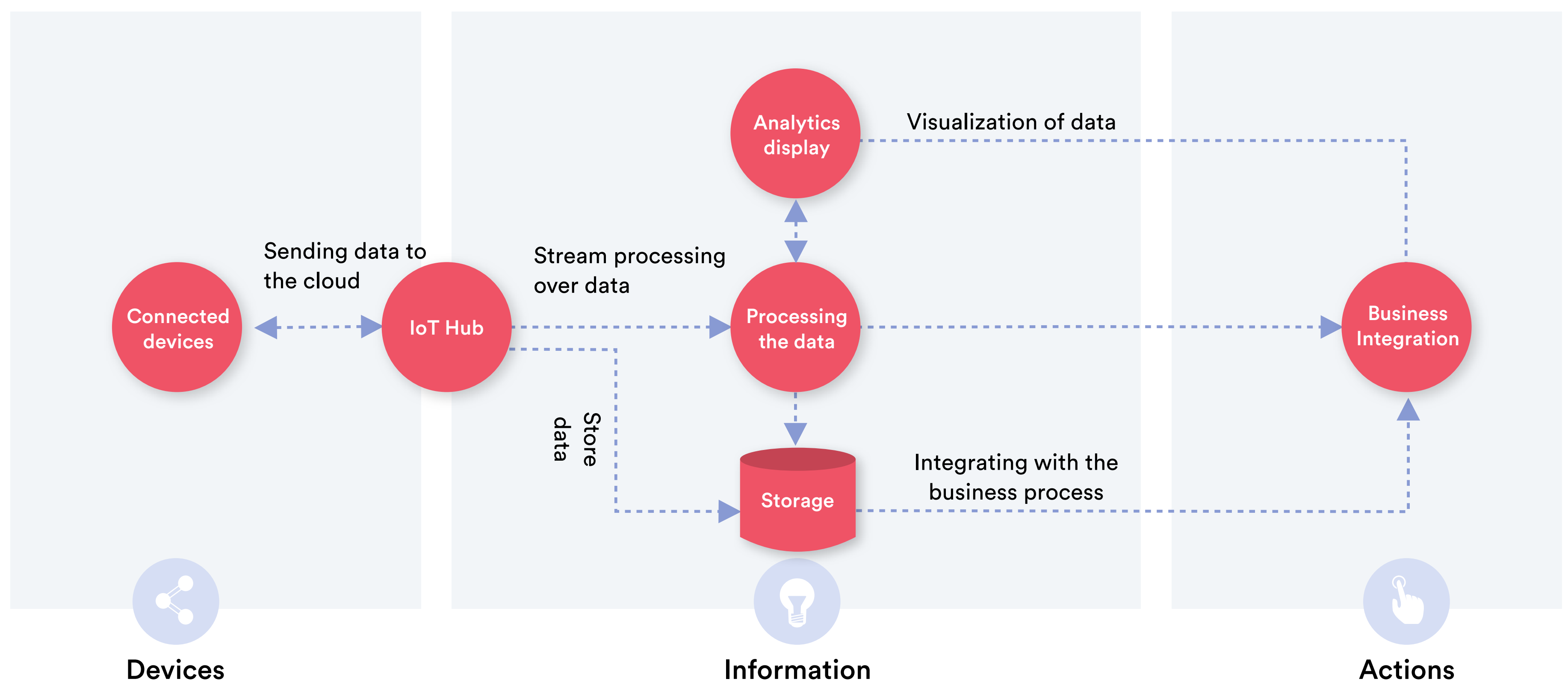
IoT solutions help supply chain managers organize their job through an improved version of the technology in use. And instead of focusing just on data tracking or barcode scanning, **Simform** aims to provide IoT-enabled software services to bring precision to your business.

As modern software solutions improve safety and security of assets, they should also shield the businesses from in-house safety threats and associated expenses. For example, IoT related **software solutions** may require you to pay for the sensors embedded in your system, and that might increase the overall budget. In such a case, ensuring that you're working with a trusted external vendor who provides quality software development is crucial.

We build custom IoT solutions that scale the software capabilities to get data in real-time helping depot managers swiftly check the stock quantities. They also come in handy when fleet managers want to allocate assets based on the in-stock and out-stock amounts and priority-based setting of deliveries. It's challenging to give fleets manual work since fleet admins or workers in the warehouse need to monitor the stocks continually, contact the supply chain markets, stock assistants, laborers, and so on. Such manual work is automated with this technology, eliminating error-prone situations and increasing productivity together with overall business efficiency and growth.

Technologies Used for IoT-Based Fleet Solutions

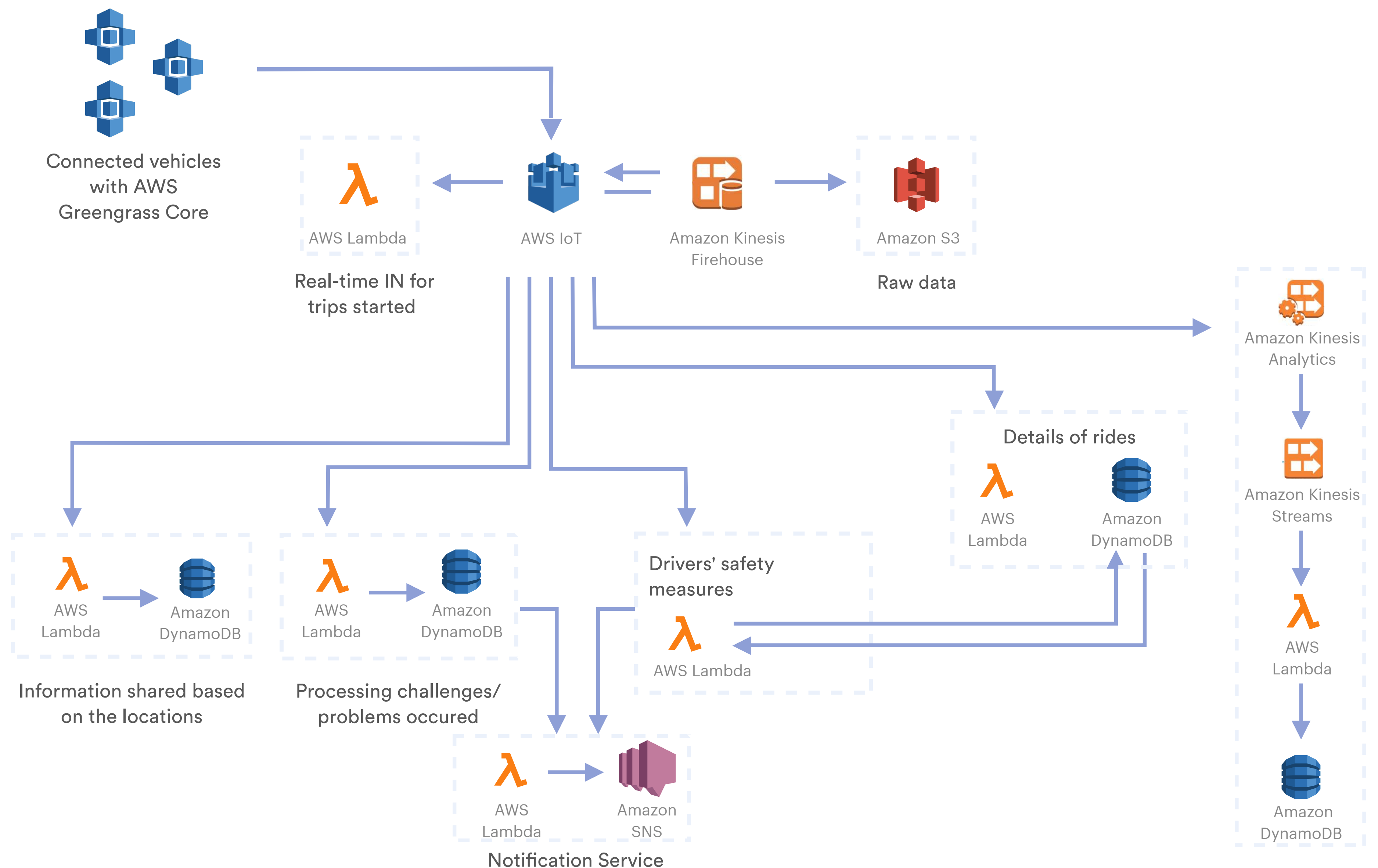
The graphical representation of IoT-application architecture below illustrates how IoT-integrated apps function, process data, and represent it in the UI layer. This is how the tech dictates the behavior of connected vehicles.



The architecture involves IoT devices exchanging information with the cloud gateways, processing it into valuable business insights, and integrating with the final representation of the information such as the dashboard or different user interfaces.

Imagine you're getting notifications on your smartwatch. This smartwatch works as an IoT device – it uses the internet through your mobile device, connects to the service provider, fetches information from the phone, and eventually relays it on the watch display.

In the connected-vehicles system, the serverless IoT architecture like AWS Greengrass syncs the data management among connected devices/vehicles, securely operates communication with other devices and vehicles, and executes the prediction-based implementations. This syncing and managing of data from all connected vehicles allows for separate operations such as acquiring trip data, ensuring driver's safety functions, sending notifications, and more.



These connected-vehicles fleet solutions will be used mostly in all the vehicle assets on the move be it trucks, ships, or any form of cargo. The integration IoT system enables communication between vehicles. As a result, you could avoid a collision in parking lots and unpredictable accidents, save time, make sure that vehicles reaching their destinations before time notify the ones arriving later about any specific snag or condition along the way, and a whole lot more.

At **Simform**, we engineer software products based on low-energy Bluetooth technology, radio protocols, WiFi-Direct, Radio Frequency Identification, and other technologies and sensors that comply with the Internet of things requirements and fulfill the needs of customizing intelligent software.

Types of sensors used in building IoT-Cloud based apps:

- Temperature and humidity sensors
- Proximity sensors
- Gas sensors
- AI and ML-powered sensors
- Level sensors

The role of different sensors is crucial in conducting any IoT, GPS or AI-based software building activity. Aside from the hardware configuration, the software is the brain working behind the curtain. It contains the main business logic to run successful fleet operations, transfer data to devices, collect and analyze the data, reflect the accurate state of the entities, and more.

These high-level fleet management software customizations work well thanks to powerful backends technology languages such as **Python, Javascript, Java, RoR**, and frameworks such as **Django**, and **Nodejs**.

Frontend frameworks like Angular also play an essential role as a building block of intelligible user interfaces. Creating these systems isn't as challenging a task as converting them into user-accessible tools is. Technologies like **Angular** and **React** are the assured pillars of this easy-to-understand UI.

For IoT connectivity, we use robust, agile, and scalable cloud-computing solutions such as IBM Watson and AWS IoT as per the project requirements. These tools are excellent in providing end-to-end solutions to scalability, connecting data and artificial intelligence programming, and supporting machine learning along with multi-cloud facilities.

Furthermore, custom fleet solutions require robust integrations of application programming interface(APIs.) Software developers or back-end engineers develop a connection between a wireless device and a service provider to exchange data among the device, cloud, and wireless internet connection.

APIs are the data pipes that transfer data and connect two intangible software-programming instances. Any web application using web sockets and web APIs can use the data to make its subsequent program.

The functions of IoT based APIs include connecting with the cloud, linking to the data interface and functions interface, communicating with the sensors, and executing tasks that involve high-level security risks, data transferring and fetching, and other critical functions.



Predictive Maintenance

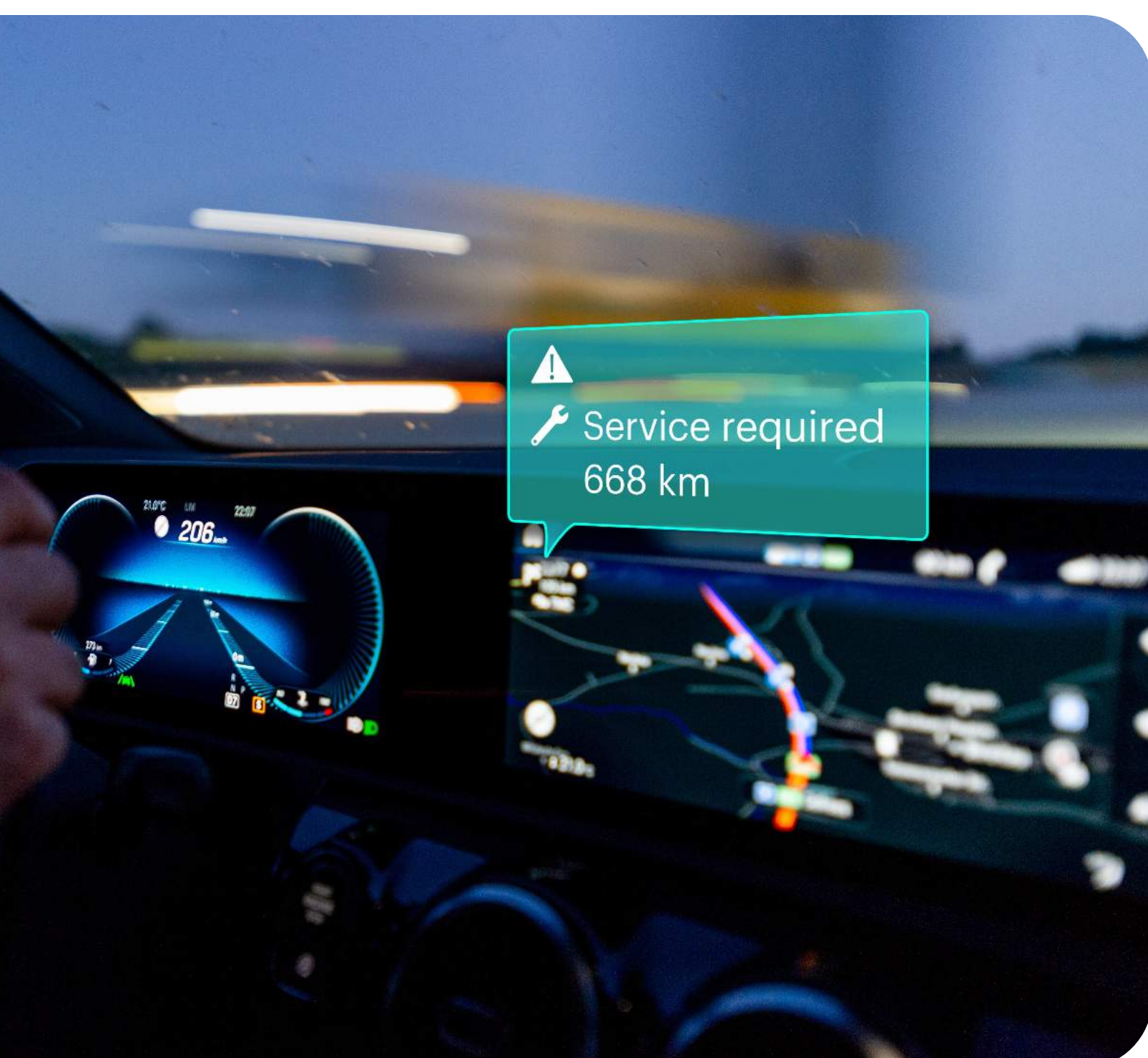
Fleet businesses face critical challenges and immediate requirements of equipment at any instance of time. Because these businesses have multiple cargos on the road at various locations and with numerous transporting materials, they could range from small to larger carriers hauling solid or liquid forms of goods. The range covers appliances, manufacturing machines, electronic devices, brand-new automobiles, and whatnot. Maintenance becomes their daily need since different freights require different approaches, tooling, and operational services.

No doubt then, maintenance is one of the core necessities of the transportation and logistics industry.

It can be done two ways: Corrective Maintenance & Preventive Maintenance.

What is corrective maintenance?

Corrective maintenance is when you act on repairing parts upon their failure. You get the repair services when parts of the vehicle are not performing ideally or create obstacles in driving and hampering proper functioning. So if you want the drivers to make deliveries on time, these repairs need to be made at once. The drawback of this approach is that it causes business downtime, increases labor, and involves more workforce, costs, and time. It also negatively impacts the overall business performance and affects the brand/business image.



What is preventive maintenance?

Preventive maintenance is to proactively solve the problems before they arise. It includes monitoring and diagnosing the health of vehicles and predicting issues that can occur in the near or longer future. It aims to optimize business costs, reduce catastrophic failures and downtime, and maintain productivity by making deliveries on time.

This approach is a form of predictive maintenance and analysis. You can achieve predictive maintenance in your fleet-management business by installing an end-to-end software solution that covers all aspects of your business. It includes fleets, types of caravans, transportation locations, number of carriers, business partners, and other important considerations. The software solution should be able to handle concurrent requests, remote diagnostics, high-speed data, third-party integrations, and IoT connectivity.

IoT-based software solutions are built using cloud-native, serverless, and microservices architecture since these apps deal with massive data, functions, and business processes. The microservices architecture helps create a less bug-prone application and speeds up the development process owing to its discrete nature. It allows us to separate the essential modules and deploy them while other modules are still being developed.

Predictive or preventive maintenance helps you schedule everything in advance. It entails various business benefits that result in growth, rapid processes, ease of work, and better human-resources management.

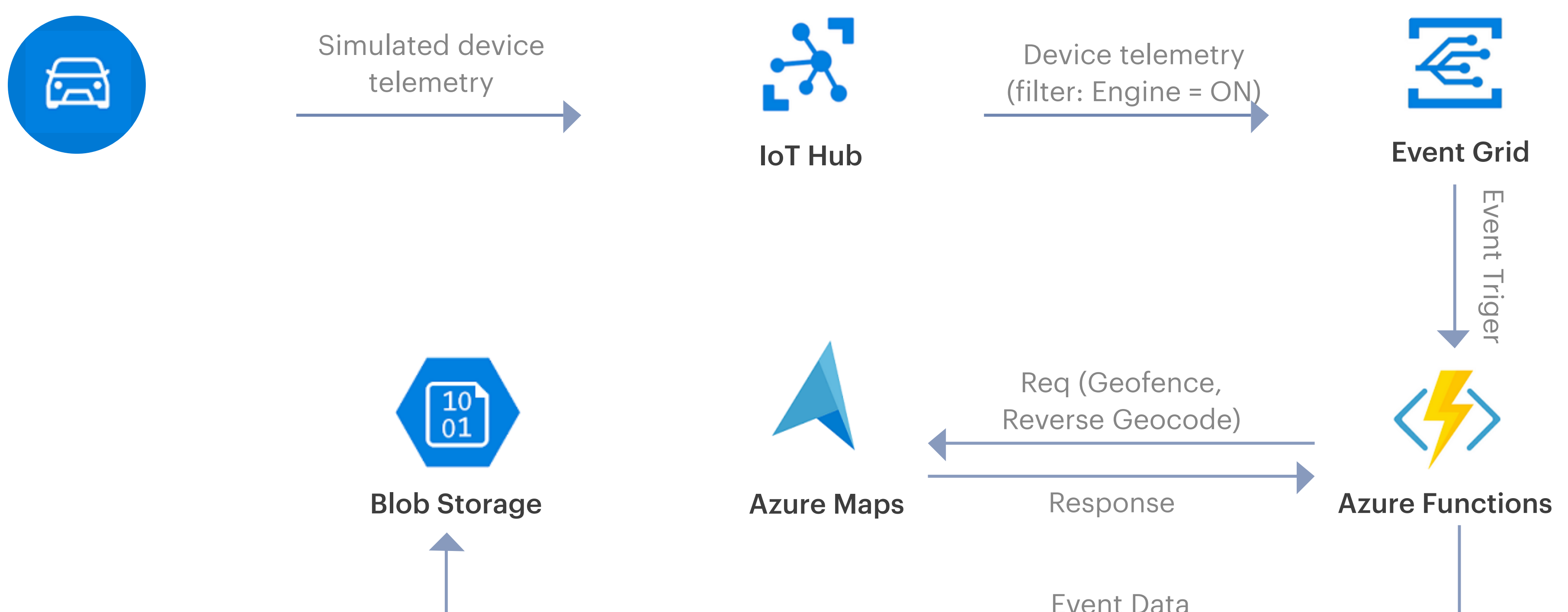
Certain developments in fleet business using preventive maintenance:

- Recognizes future problems in different parts of the vehicle increasing its overall efficiency and lifespan and that of its parts
- Frequent checks of sensitive information such as engine performance, tire pressure, hydraulics, hopper camera
- Instant notifications/alerts to the immediate responders and drivers using the over-the-air technology, GPS systems, Internet of things, and customized functionality as per the fleet needs
- Real-time updates, sensor-based information
- Better decision making leading to higher profits thanks to the immediate information shared among connected vehicles
- Reduced or zero vehicle downtime
- Reduced labor and maintenance costs
- Enhanced safety features

Spatial Operations and Routing

The primary function of the Internet of things is to analyze the data connecting different resources that lead to useful, actionable insights. And spatial operations and routing is a critical phase of automating fleet operations and business functions. Spatial operations are processes that use geographic information. They process this information using geometry functions and give commands/signals in real-world scenarios. IoT hubs can transmit data to the connected devices and produce the output using spatial operations and geofencing calculations.

For example, the diagram below shows the exchange of signals between a simulated device and the IoT hub. A serverless hub like Azure processes the information and generates the final output.



Source : <https://docs.microsoft.com/en-us/azure/azure-maps/tutorial-iot-hub-maps>

It could be anything related to the transportation system – a fleet carrier, ride-sharing cars, or special-cargo marine ships – routing and spatial operations play a crucial role in solving all fleet-management problems.

Take a ride-sharing system like Uber. The geofencing captured by the Uber maps informs drivers of the location from where the maximum number of passengers book a ride. The rider can see Uber drivers in the vicinity once they enter the location. With the help of a global positioning system, satellites fetch the coordinates, accurately capture the space, and create a geofence.

The same principle applies to fleet operations. Managers can learn the details from their dashboard about the requirements in a particular area, location of the specific freight carrier, and more.

The software system enables users to capture and interpret the spatial data, locate the actual data, prepare the readable format, and understand the map layers. It also uncovers unrecognizable geographical patterns and facts about business activities.

Examples of spatial operations in fleet management

Geofencing

To successfully implement a group of operations and communications, you need an API/programming algorithm that regularly captures the coordinates and works as a communication pipe carrying data. For geofencing to be possible, we need to retrieve the proximity of coordinates, pinpoint the exact location, convert data into proper arrangement from the pure-coordinate system, and locate a target object.

Geofencing is one such example of spatial operations that works to locate objects and an area on the earth for various purposes. It includes tracking of assets, detecting moving objects, monitoring idle fleets, and the like.

For example, a remote dashboard showing accurate contours of parked vehicles/fleets. Or security forces being able to locate the exact location of an accident that occurred in a remote area.

Closest Point

The closest point is a reference of point to find the nearest location.

For example, a freight carrier needs to find the nearest petrol station. Or maybe another fleet carrier from the same company. The software application should produce accurate results by processing the algorithm with a given set of points via user-uploaded data and other data-uploading APIs.



Examples of Routing operations in FM

Route direction

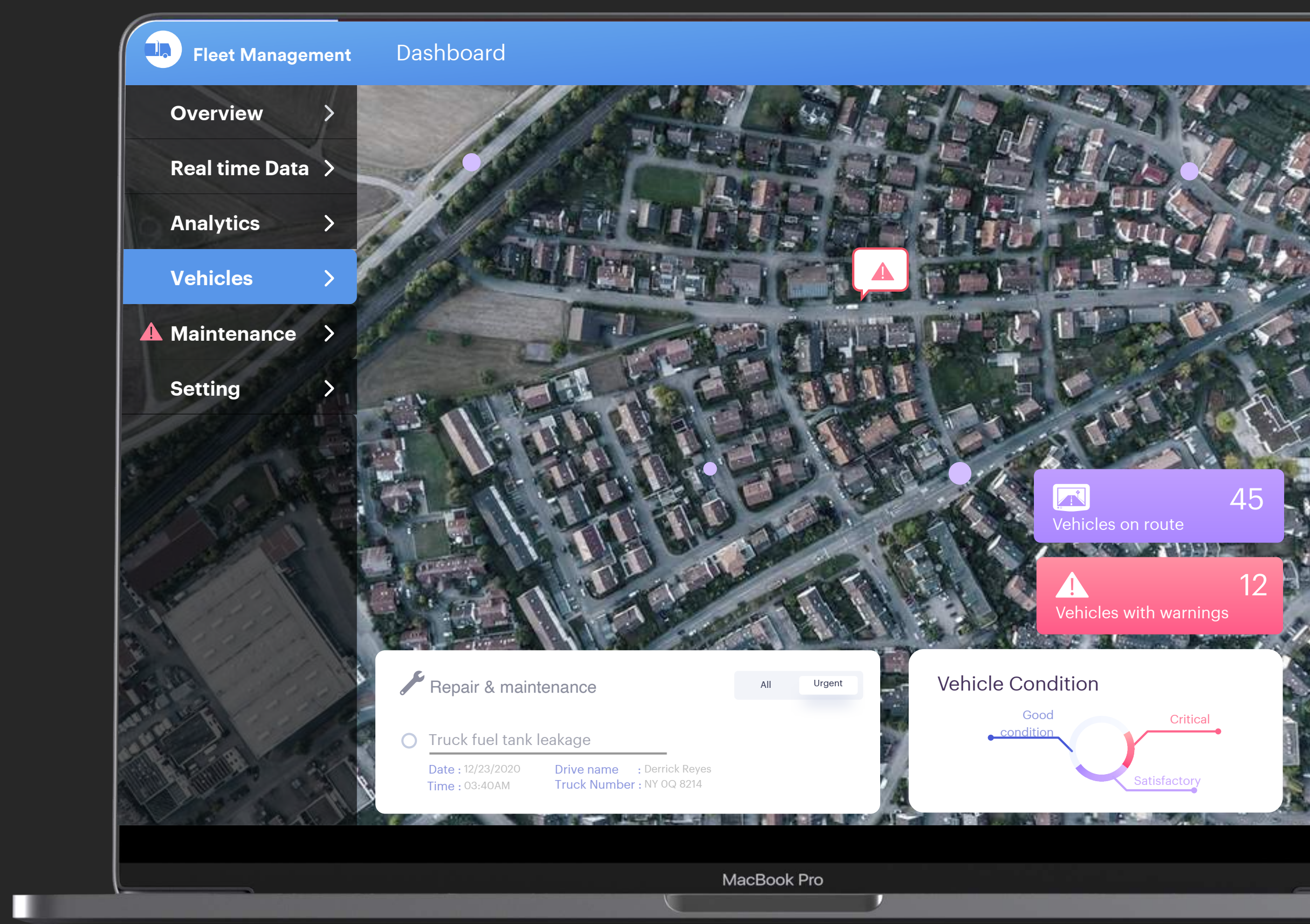
It's a measure between an origin and destination. It includes various aspects such as peak hours of traffic during the day, time required to reach a destination, conditions of the roads, any obstacles along the way, calculating approximate speed on certain roads, and much more. For freight carriers to make deliveries on time, this proactive route-optimization system helps decrease the time of delivery. It uses spatial information to analyze actual data and generate a final notification for drivers.

Isochrone

A type of spatial operation that considers the number of locations to be visited by the vehicle. It then calculates possible conditions based on fuel, time, distance, and budget requirements. It's chiefly used to search inside a boundary defined between locations. Clearly, an Isochrone map is very useful for transportation planning.

Remote monitoring and Dashboard

The remote dashboard is a set of organized data from all your business activities. The real-time and historical representation of data in a single screen is readily available for analytics to explain its business implications. You can modify the key performance indicators of your business, view and filter desired data, and configure settings according to the business requirements.



Which information is managed by a remote dashboard?

- Data based on business intelligence for future decisions (predictive analysis)
- Effective decision-making factors
- Optimization suggestions (for example, optimized routes)
- Trips and destinations
- Information related to expenses (fuel, trips, performance metrics, idle costs, service repairs, etc.)

Features:

- Vehicle's services & diagnostics planner
- Real-time data of vehicle/freight carrier such as location, status, alerts
- Predictive analytics of coverall cost, fuel consumption



Device stats

Reads different metrics measured by the device installed in a vehicle, displays various statistics such as the temperature of the vehicle, weather conditions, traffic scenarios, estimated trip time, etc



Device locations

Represents a device's current location in real-time, its previous locations, location-related information such as date and time of the trip, time required to complete the trip, and idle time at a particular location



Alerts

Shows necessary alerts and warnings related to vehicle service or immediate repairs, fuel consumption, tire pressure, vehicle health, battery life, etc. under the key indicators



Telemetry

Real-time indication of all fundamental telematics including vehicle speed, idle time, location details and pictures through rear & front camera, etc.



Analytic

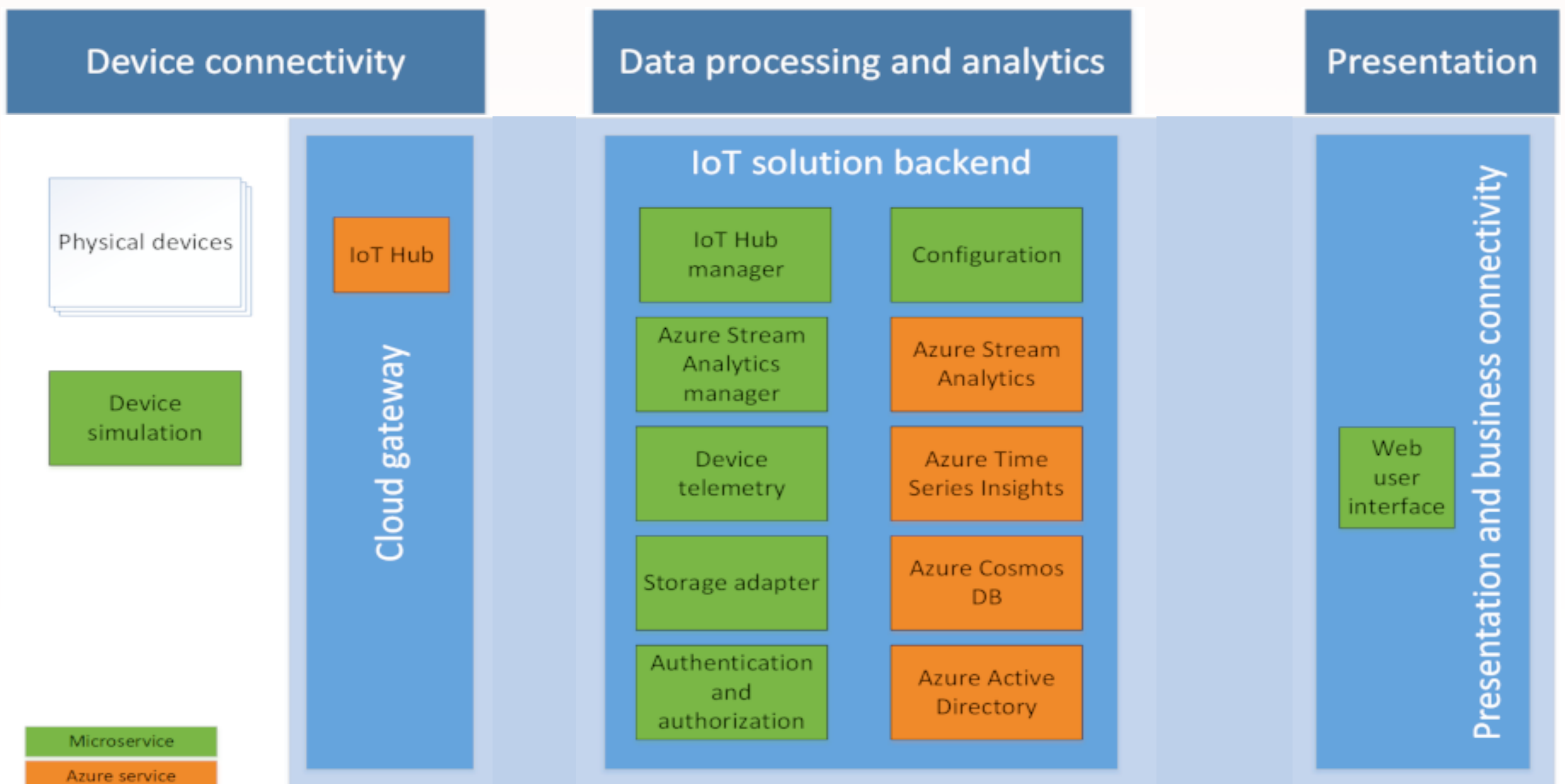
Calculation of ideal time for a vehicle to make trips, during idle state, traffic analysis based on the previous data for a particular time of the day and location, fuel consumption, best practices to avoid unpredictable circumstances

The logical architecture of Remote Dashboard

Logical architecture is the graphical representation of backend components in a remote dashboard. It has three separate segments for device connectivity, data processing and analytics, and final presentation for business insights. The unit for data processing and analytics is the core segment for producing readable real-time data. In contrast, device connectivity links the physical devices to cloud gateways and IoT solutions.

The final representation of data has a huge backend behind it which collects information from various hardware components like sensors, processes those raw inputs using the spatial geometry, and relays them onto the remote dashboard.

We've taken Microsoft's Azure remote-monitoring backend architecture as a reference.



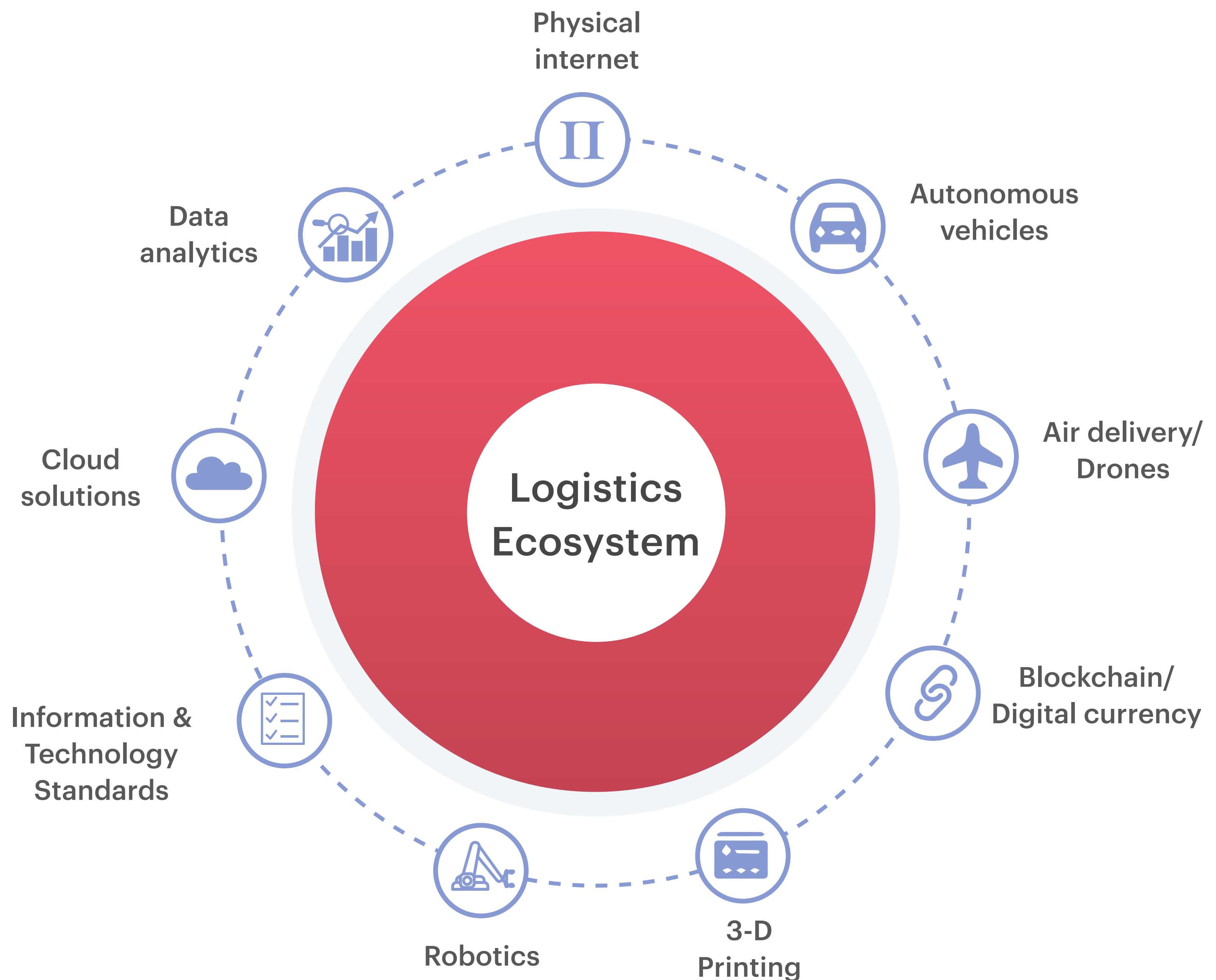
Source : <https://docs.microsoft.com/en-us/azure/iot-accelerators/iot-accelerators-remote-monitoring-sample-walkthrough>

Augmented and Virtual Reality

You must be aware of augmented-reality and virtual-reality technologies and may feel that they are not part of the real-world solutions yet. However, augmented and virtual reality are great tech stack to solve present-day challenges.

Augmented reality superimposes the graphics on real-world objects and creates an enhanced experience for users. It's an interaction carried out via graphics, computer-generated sensors, audio, and video. Virtual reality, on the other hand, is a simulated environment. Users get to view objects or scenarios in three-dimensional reality using devices like cardboards or Google glasses. Both these computer-generated graphical representations are beneficial when it comes to fleet management.

For example, as a fleet manager, you want useful data like real-time traffic updates, temperature, and alerts onto a display that minimizes a driver's distraction. Or imagine your dispatch hub that receives goods in bulk has a labor staff collecting information about each parcel. You can employ wearable devices that gain important information on parcels such as weight, delivery address, contents, etc. It'd minimize manual tasks, improve the focus of drivers, streamline the process, and save a lot of handling damages.



While AR is already changing the way it deals with vehicles via its augmented widescreens, many industries and vehicle manufacturers have opted for integrated virtual-reality solutions. It also includes displaying information that contains obstacle detection, traffic alerts, video surveillance, and more using VR-based software solutions. VR and AR can also communicate to riders the parallel conditions of other vehicles. It means you can monitor the speed of other riders, proactively spot possible collisions and blind-spots, and access many other useful features. Even for maintenance and service repairs, you can employ smart glasses with step-wise information related to repairing a certain part of the vehicle. This is especially useful in remote locations where drivers aren't within reach of any service center and have to make repairs themselves.

Warehouse management gets even more streamlined with these technologies. Locating items using object-recognition software ensures the accuracy of picking specific items, hence improving the ultimate customer experience. So the future for solutions based in augmented and virtual reality is bright, and the fleet and transport industries are set to earn tremendous benefits from these advancements.

If you are wondering how to choose the best telematics solution for fleet management of your business, then here is a list of important questions to keep on your checklist.

- ✓ What are the critical touchpoints in your business that need a telematics solution?
- ✓ Which is the best technology in telematics for fleet management that will best suit your business?
- ✓ What are the prerequisites to integrate a telematics solution?
- ✓ What are the added benefits/features that such a solution will bring to your business?
- ✓ Will it particularly improve business operations and reduces costs?
- ✓ How will you scale the solution across business processes for better fleet management?
- ✓ What is the tech-stack requirement to execute a telematics solution for your fleet management?
- ✓ How to develop such a tech stack within your budget, and suitable for your fleet management?

Once you find answers to these questions, there will be a better understanding of the telematics solution for your business. But, if you have specific needs that need personalized solutions, you need to create a digital ecosystem. Such an ecosystem will include all the tech stack for warehouses, fleet maintenance, tracking of shipments, remote monitoring, and even transportation efficiency.

As you build an ecosystem of customized solutions, the cost can increase or decrease depending on the scale of the solution. But with answers to all your questions in the checklist, you will have an idea about approximate pricing.

The Know-how to the fleet management solution

With this ebook, we have tried to explore the latest advancements in telematics solutions for fleet management. We hope this helps you understand the type of tools and tech-stack needed to develop an advanced fleet management solution. Or what to choose for your business and how it can improve the efficiency of operations. If you want to execute a telematics solution and want to know more about the cost of the technology stack then feel free to get in touch with our professionals.

Concluding remarks...

This is not a closed report,
It's an open conversation.

With this report, we hope to initiate a conversation and empower you to have a meaningful conversation around test automation backed with data and insights.

If you have any questions on the topic, we'd be happy to hear from you. Send me an email hiren@simformlabs.com or connect with me on **Twitter** or **LinkedIn**.



Hiren Dhaduk

VP of Technology

hiren@simformlabs.com



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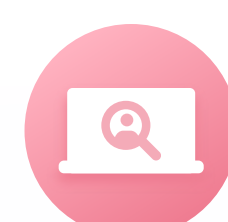
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